Code.org: Debugging Tips

Every coder gets bugs. The best ones just fix them faster!

Code.org | Beginner | Ages 6–14

💡 The Big Idea

- A bug is a mistake in your code that makes it do something different from what you planned.
- Debugging means finding the bug and fixing it.
- Every programmer, even the experts, writes bugs. Fixing them is normal.
- Stay calm and be curious. A bug is a puzzle to solve.

🔍 The 5 Step Debug Plan

- Step 1: Say what you wanted to happen.
- Step 2: Run the program and watch what really happens.
- Step 3: Find the first place where the two are different.
- Step 4: Change one thing only.
- Step 5: Run it again and check. Repeat if needed.

🪲 Bugs to Look For

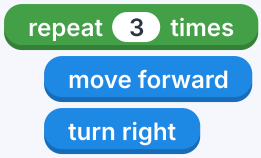
- Wrong order: the steps are in a different order from what you need.
- Wrong number: a loop repeats too few or too many times.
- Wrong direction: turn left instead of turn right.
- Missing block: you forgot a step like move forward.
- Wrong event: the code starts at the wrong time.

🧰 Helpful Tricks

- Slow down. Use the Step button if the puzzle has one.
- Say each block out loud as if you are the character.
- Use the Hint button, then try again yourself.
- Act it out. Walk the steps with your own body.
- Take a short break and look again with fresh eyes.

💻 Bug and Fix Examples

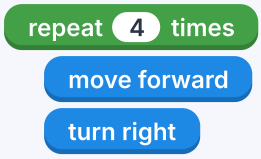
Bug: the square is not finished — Code.org blocks



▶ What happens:

The character draws only three sides of a square. The shape is open.

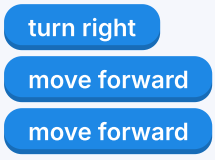
Fix: change the loop count — Code.org blocks



▶ What happens:

Four sides are drawn, so the square is complete.

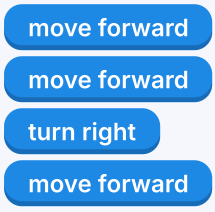
Bug: wrong order — Code.org blocks



▶ What happens:

The character turns first and then walks the wrong way. It misses the goal.

Fix: swap the steps — Code.org blocks



▶ What happens:

The character walks to the corner, turns, and reaches the goal.

📚 Key Terms

Bug
A mistake in code.

Debugging
Finding and fixing bugs.

Test
Run the code to check it works.

Step
Run the code one block at a time.

Hint
Help shown by the puzzle.

Loop count
How many times a loop repeats.

Sequence bug
The steps are in the wrong order.

Prediction
What you think the code will do before you run it.

Error
A message that says something went wrong.

Persistence
Not giving up when it is hard.

🧮 Debug Checklist

1. What should happen?
2. What actually happens?
3. Where do they first differ?
4. Change ONE thing
5. Run and test again

✏️ Worked Examples

Q: The character walks past the goal. What could be wrong?

A: The loop probably repeats too many times, or a move forward block is extra. Count the steps and compare.

Q: The character turns the wrong way at a corner.

A: The turn block may be left instead of right. Check each turn block against the map.

Q: Nothing happens when you press Run.

A: There may be no starting block, or the blocks are not connected. Check that everything is attached to the start block.

🚫 Common Mistakes

- Changing many blocks at once, so you do not know what fixed it.
- Giving up instead of using Hint or Step.
- Not predicting the result before running.
- Blaming the computer. Usually the order or a number is wrong.
- Deleting everything and starting again, when only one block was wrong.

👨‍👧 Parent Corner

- Ask: What did you expect? What happened? Do not fix it for them.
- Say bugs are normal. Share a mistake you fixed at work.
- Praise the process: Great job staying calm and testing.
- Let them teach you why the bug happened.
- Take a break if frustration rises, and come back later.

🎯 Quick Check

1. What is a bug?

✅ A mistake in a program.

2. What does debugging mean?

✅ Finding and fixing bugs.

3. Why change only one thing at a time?

✅ So you know which change fixed or caused the problem.

4. A square has only 3 sides. What is the likely bug?

✅ The loop repeats 3 times instead of 4.

5. What is a good first step when a program does not work?

✅ Say what you wanted, then watch what really happens.